

Computational Physics 2 Classes No.3 - Report

Hamiltonian eigenstates for a 2D square infinite quantum well

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1 Introduction

In this report, the stationary states of a particle confined in a two-dimensional infinite square quantum well are studied. The problem is solved numerically with the finite element method in the Galerkin formulation, which leads to a generalized eigenvalue problem for the Hamiltonian. The obtained energy eigenvalues and eigenfunctions are compared with the exact analytical results, and the convergence of the spectrum is analyzed for bilinear and biparabolic basis functions.

2 Results

The numerical solution of the generalized eigenvalue problem

$$\mathbf{H}c = E\mathbf{O}c$$

was obtained for two types of basis functions: bilinear and biparabolic. The calculations were performed for several mesh densities, with

$$N = 2, 3, 4, 5, 6, 8, 10,$$

and the side length of the square well fixed at

$$L = 5.$$

Only the positive part of the spectrum was analyzed, since the negative eigenvalues are artificial states introduced by the boundary-condition procedure and correspond to the decoupled boundary nodes.

The exact energy levels for the infinite square well are

$$E_{n_x, n_y} = \frac{(n_x^2 + n_y^2)\pi^2}{2L^2},$$

therefore they provide a direct reference for assessing the quality of the numerical approximation. In the following, the first ten positive eigenvalues and several of the lowest-energy eigenfunctions are discussed.

2.1 Comparison of the numerical and exact spectra for $N = 10$

Figure 1 shows the first ten positive eigenvalues obtained for the bilinear and biparabolic bases for the mesh parameter $N = 10$, together with the exact analytical values. This plot provides the most direct comparison of the two numerical approximations at the finest mesh used in the calculations.

The numerical spectrum obtained with both basis sets reproduces the expected structure of the low-energy part of the exact spectrum. In particular, the ordering of the eigenvalues is consistent with the theoretical result for the two-dimensional square infinite well. The biparabolic basis yields visibly better agreement with the exact values than the bilinear basis, which is expected because higher-order shape functions give a more accurate approximation of the eigenstates on the same mesh.

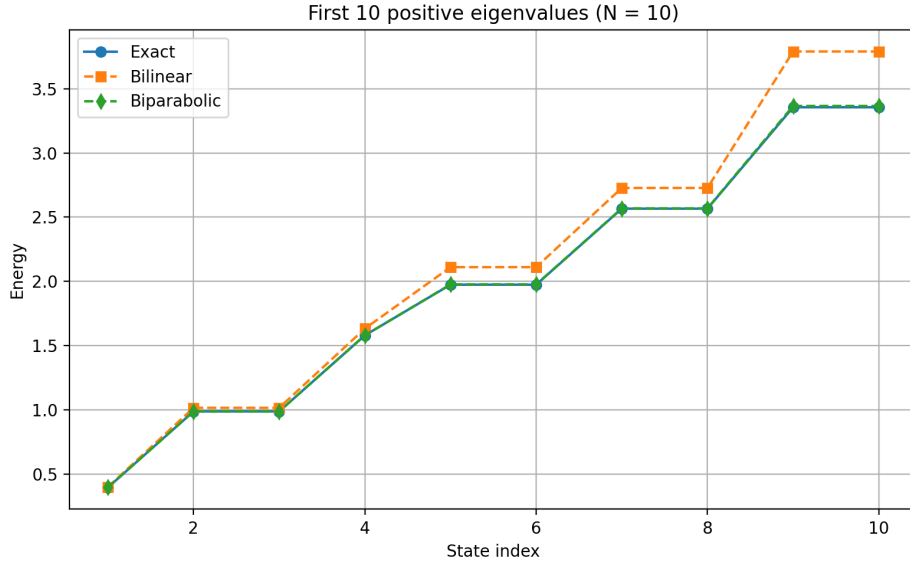


Figure 1: Comparison of the first ten positive eigenvalues for $N = 10$. The exact analytical energies are shown together with the numerical results obtained using bilinear and biparabolic basis functions.

2.2 Convergence of the eigenvalues with mesh refinement

To study convergence, the first few energy levels were computed for increasing values of N . Figures 2 and 3 show the numerical eigenvalues as functions of N for the first five states, separately for the bilinear and biparabolic bases. In each case, the corresponding exact energies are indicated by horizontal dashed lines.

For both basis sets, the numerical eigenvalues approach the exact values as the mesh is refined. This confirms that the finite element discretization converges correctly to the continuum solution of the Schrödinger equation. The convergence is systematic for all displayed low-energy states. As expected, the approximation of the lowest states is generally more accurate than that of the higher ones, because higher excited states have a more oscillatory structure and are therefore more sensitive to the finite spatial resolution of the mesh.

The biparabolic basis gives a noticeably faster convergence with N than the bilinear basis. For the same mesh density, the energies obtained with biparabolic shape functions lie closer to the exact analytical values. This indicates that the higher-order local approximation significantly improves the representation of the eigenfunctions and, consequently, the resulting eigenvalues.

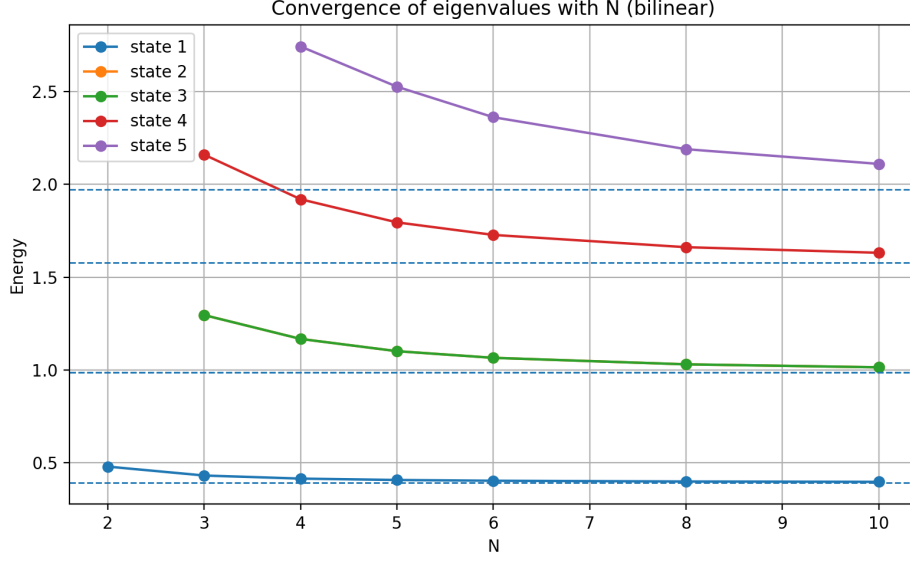


Figure 2: Convergence of the first five positive eigenvalues with increasing N for the bilinear basis. The dashed horizontal lines denote the exact analytical energies.

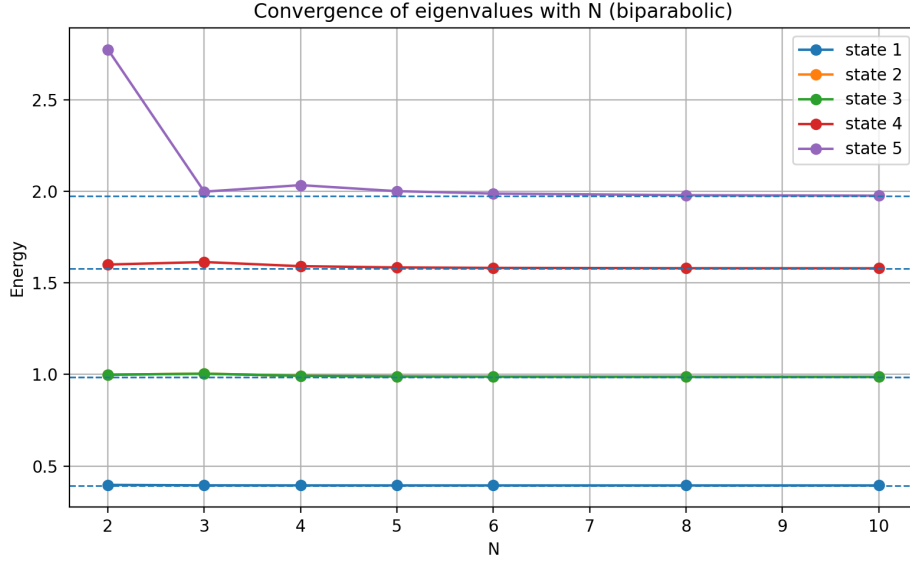


Figure 3: Convergence of the first five positive eigenvalues with increasing N for the biparabolic basis. The dashed horizontal lines denote the exact analytical energies.

2.3 Convergence of the absolute errors

A clearer picture of the numerical accuracy is given by the absolute errors

$$|E_{\text{num}} - E_{\text{exact}}|.$$

These errors are shown on a logarithmic scale in Figures 4 and 5, again for the first five states and for both types of basis functions.

The logarithmic presentation makes the convergence trend particularly transparent. In both cases, the errors decrease as N increases, which confirms the stability and correctness of the numerical method. However, the reduction of the error is much more pronounced for the biparabolic basis than

for the bilinear one. This behavior is fully consistent with the theoretical expectation that higher-order basis functions should deliver greater accuracy for smooth solutions.

It is also visible that the error is state-dependent: the lowest-energy states are reproduced most accurately, while the higher states converge somewhat more slowly. This is natural, since higher states possess shorter spatial wavelengths and require a finer discretization to be represented with comparable quality.

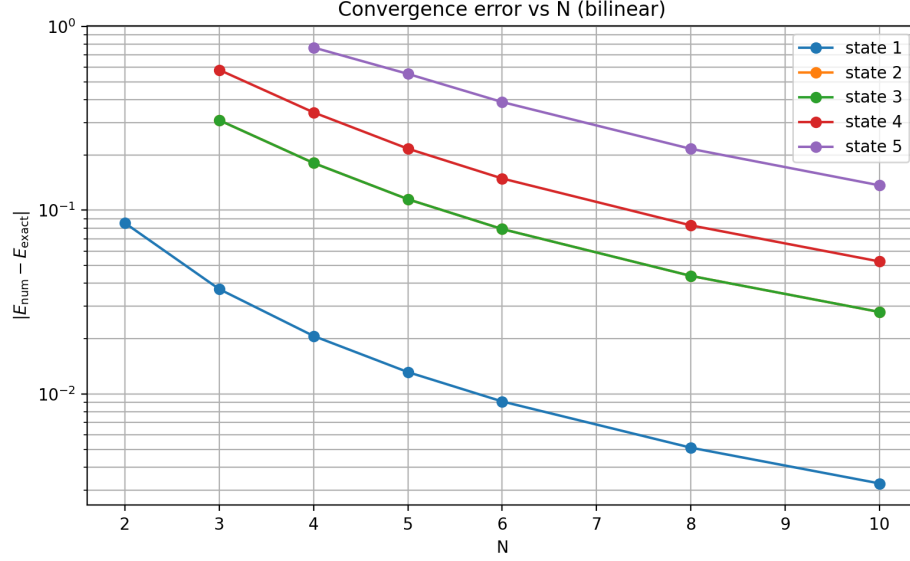


Figure 4: Absolute errors of the first five positive eigenvalues as functions of N for the bilinear basis. The vertical axis is shown on a logarithmic scale.

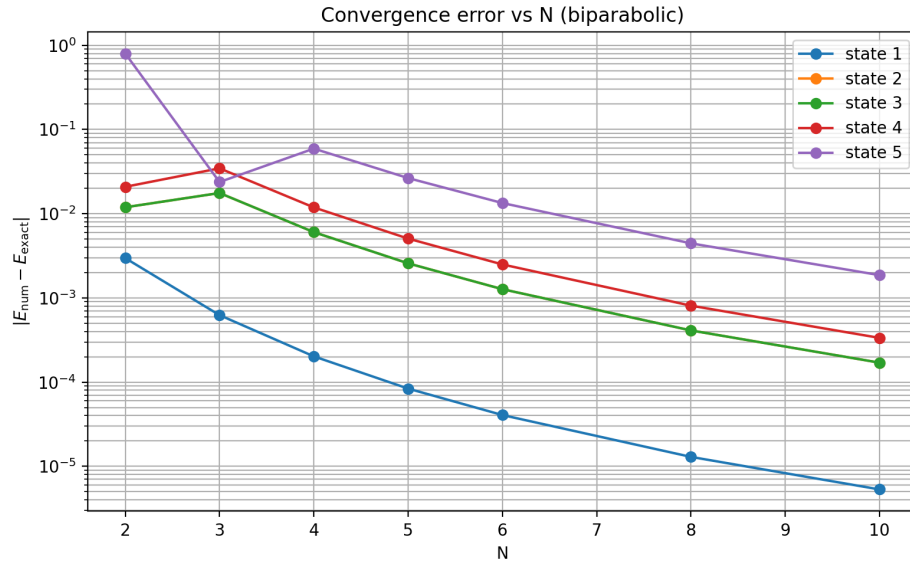


Figure 5: Absolute errors of the first five positive eigenvalues as functions of N for the bipolarabolic basis. The vertical axis is shown on a logarithmic scale.

2.4 Wave functions of the lowest-energy states

The numerical eigenvectors obtained from the generalized eigenvalue problem represent the values of the wave functions at the mesh nodes. For $N = 10$, contour plots of the first four positive eigenstates

were generated for both basis sets. These are shown in Figures 6–9 for the bilinear basis and in Figures 10–13 for the biparabolic basis.

The ground-state wave function has the expected shape: it is nonzero in the interior of the square, vanishes at the boundaries, and has no internal nodal line. The next excited states show the appearance of nodal structures characteristic of the analytical solutions $\sin(n_x\pi x/L)\sin(n_y\pi y/L)$. In particular, the first excited states are associated with one nodal line in either the x - or y -direction, corresponding to the degeneracy between the $(n_x, n_y) = (1, 2)$ and $(2, 1)$ states. Higher states exhibit more complicated nodal patterns, in agreement with the expected analytical form.

The contour plots obtained with both bases reproduce the correct qualitative behavior of the low-energy eigenfunctions. However, the biparabolic basis gives smoother and more accurate spatial profiles, especially in regions where the wave function varies more strongly. This is consistent with the improved spectral accuracy observed earlier.

Because eigenvectors are defined only up to an overall sign, the sign of a numerical eigenfunction is arbitrary. Therefore, the physically relevant information is contained in the spatial structure, nodal lines, and symmetry properties of the wave functions rather than in the absolute sign of the plotted field.

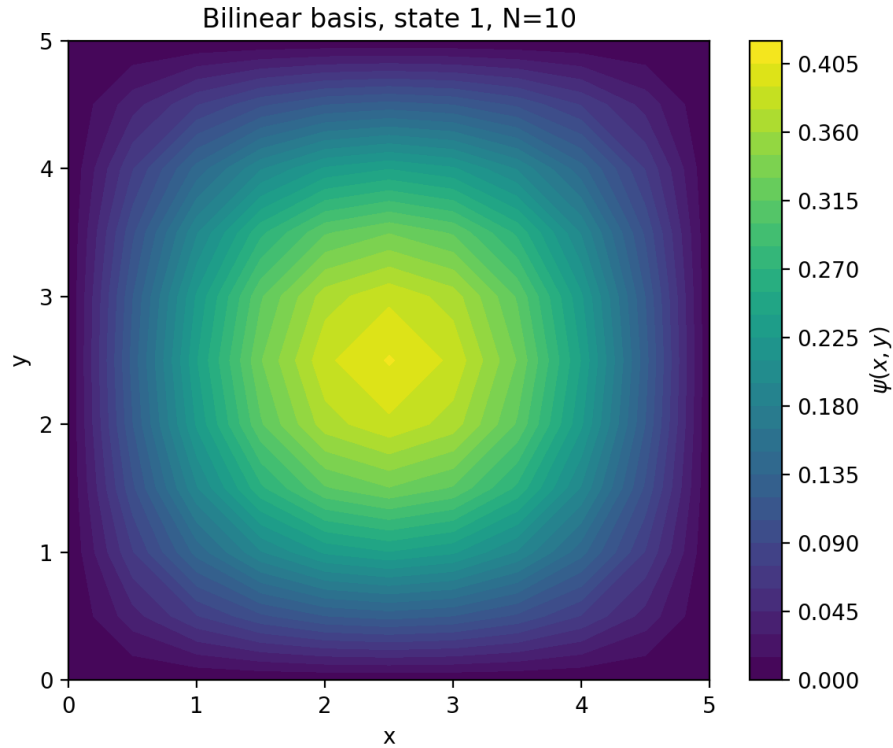


Figure 6: Wave function of the first positive eigenstate for the bilinear basis, $N = 10$.

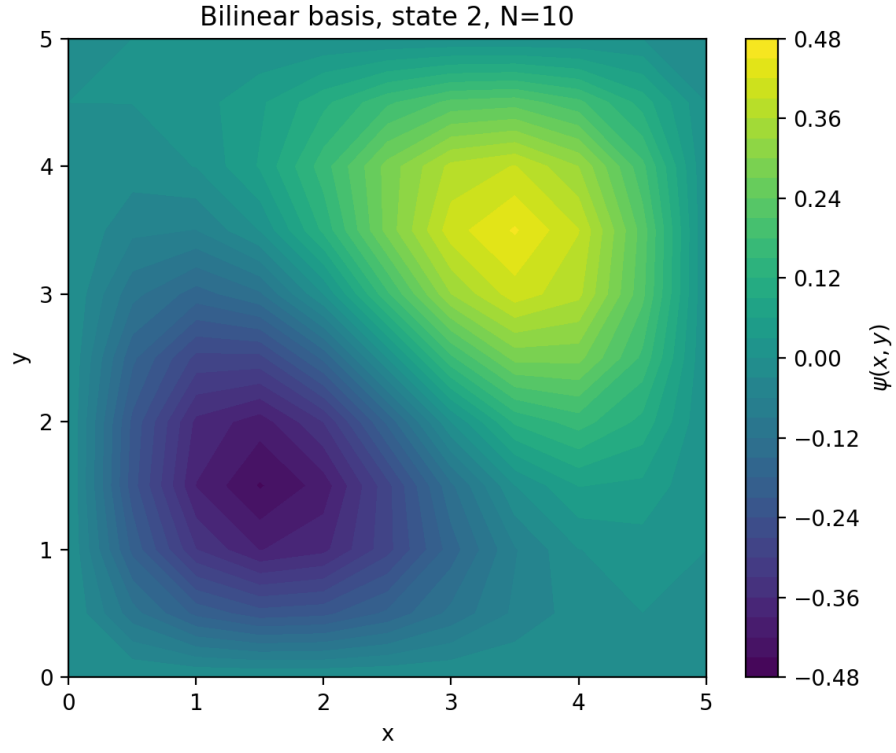


Figure 7: Wave function of the second positive eigenstate for the bilinear basis, $N = 10$.

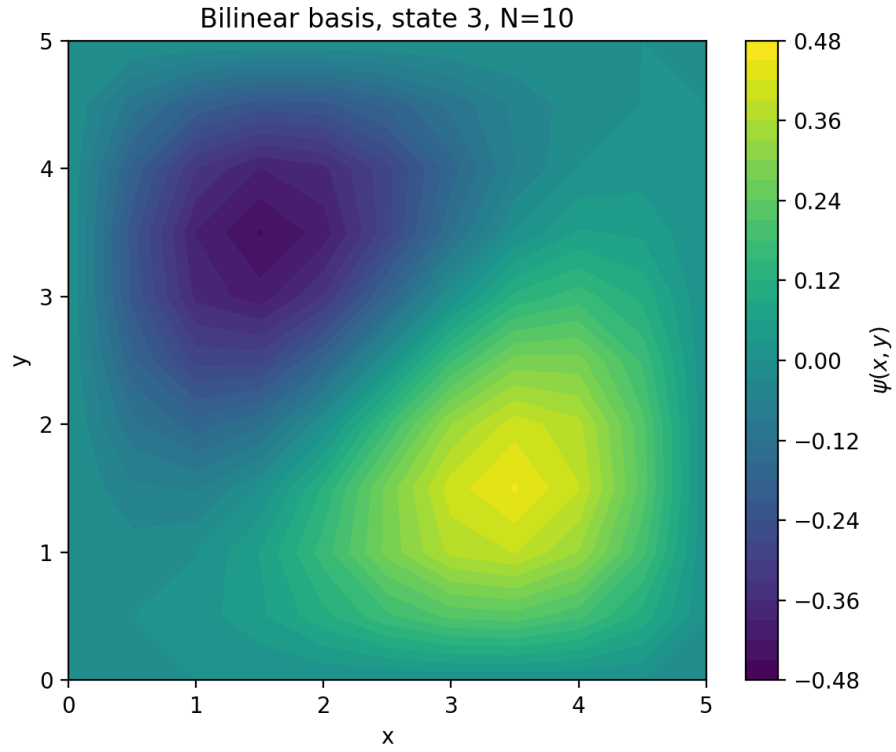


Figure 8: Wave function of the third positive eigenstate for the bilinear basis, $N = 10$.

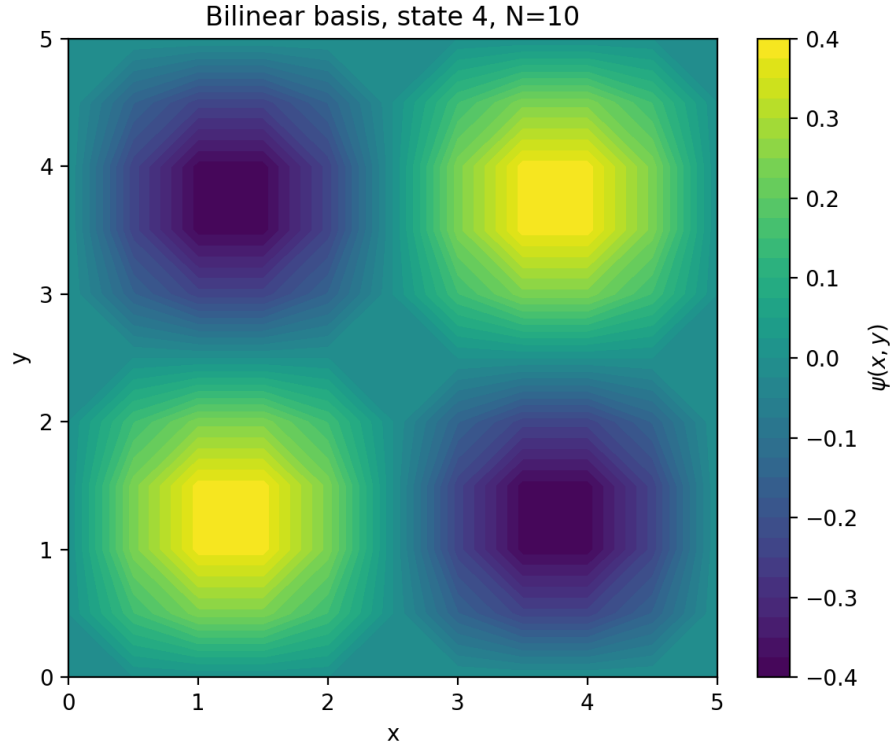


Figure 9: Wave function of the fourth positive eigenstate for the bilinear basis, $N = 10$.

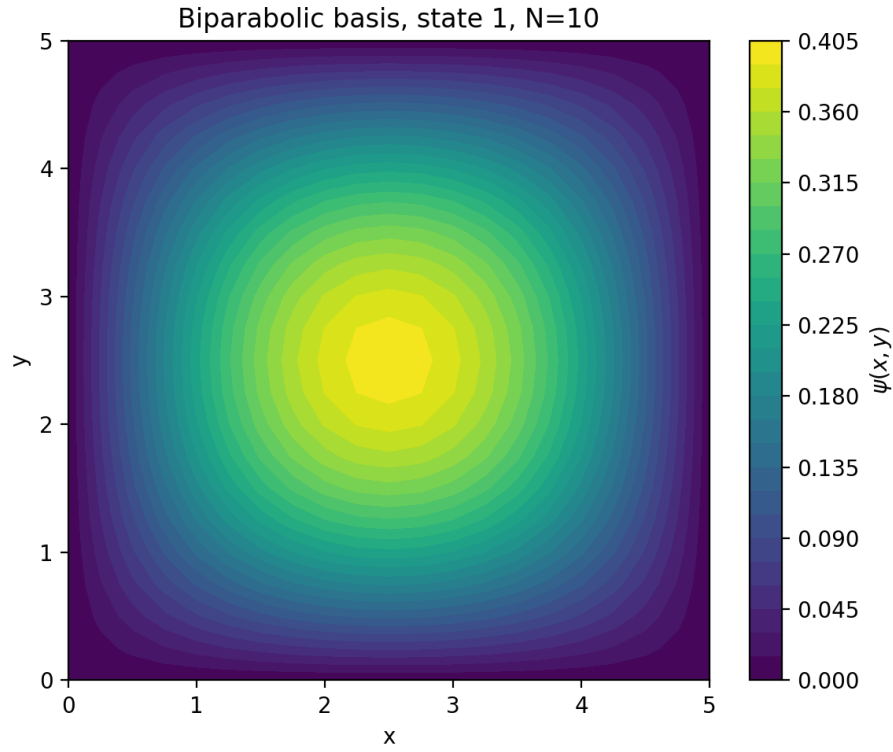


Figure 10: Wave function of the first positive eigenstate for the biparabolic basis, $N = 10$.

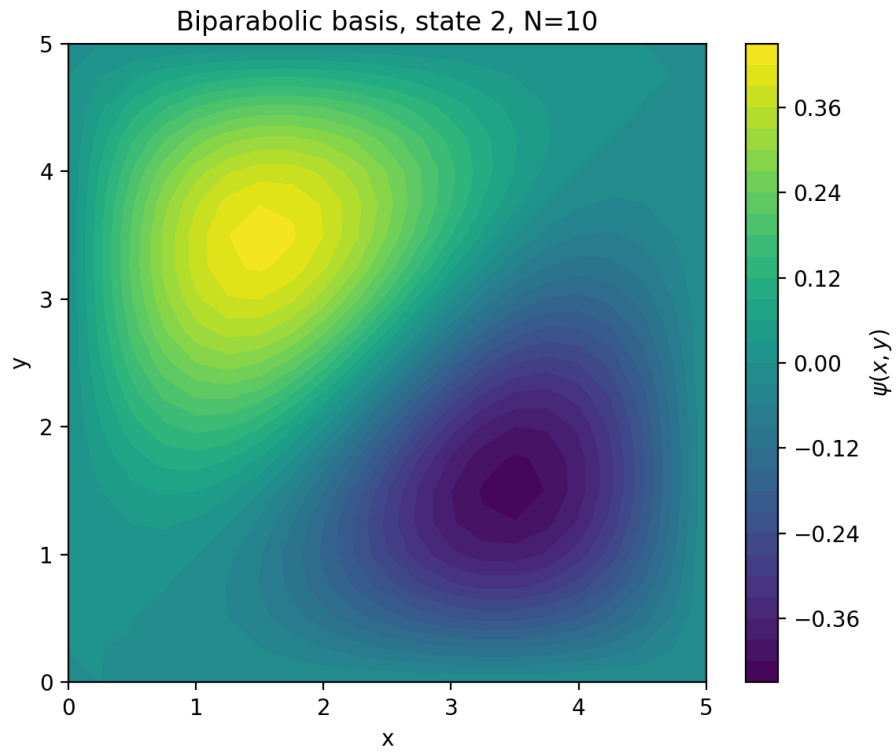


Figure 11: Wave function of the second positive eigenstate for the biparabolic basis, $N = 10$.

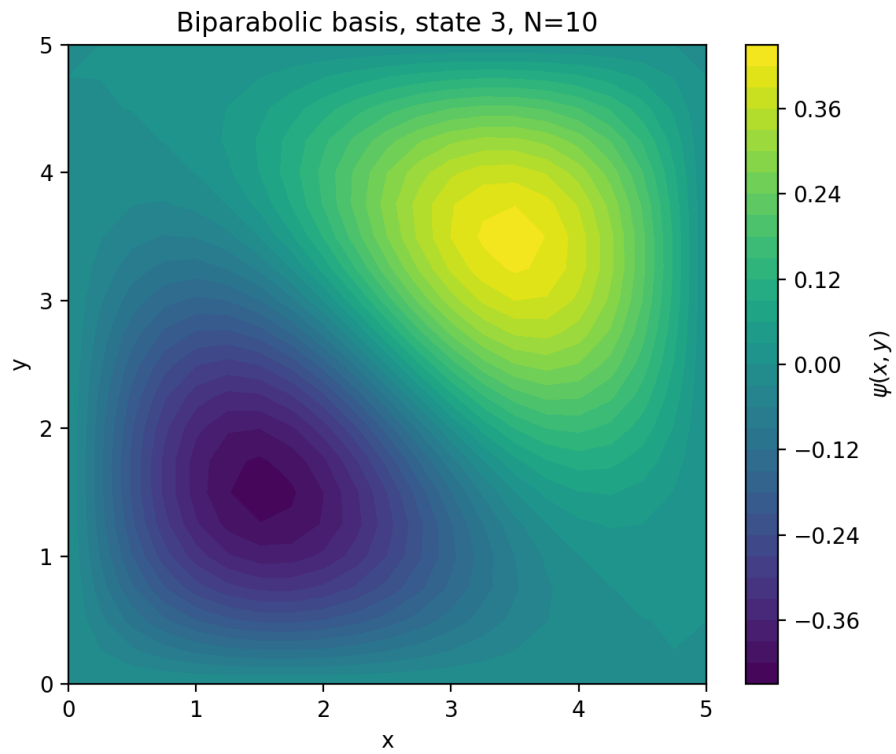


Figure 12: Wave function of the third positive eigenstate for the biparabolic basis, $N = 10$.

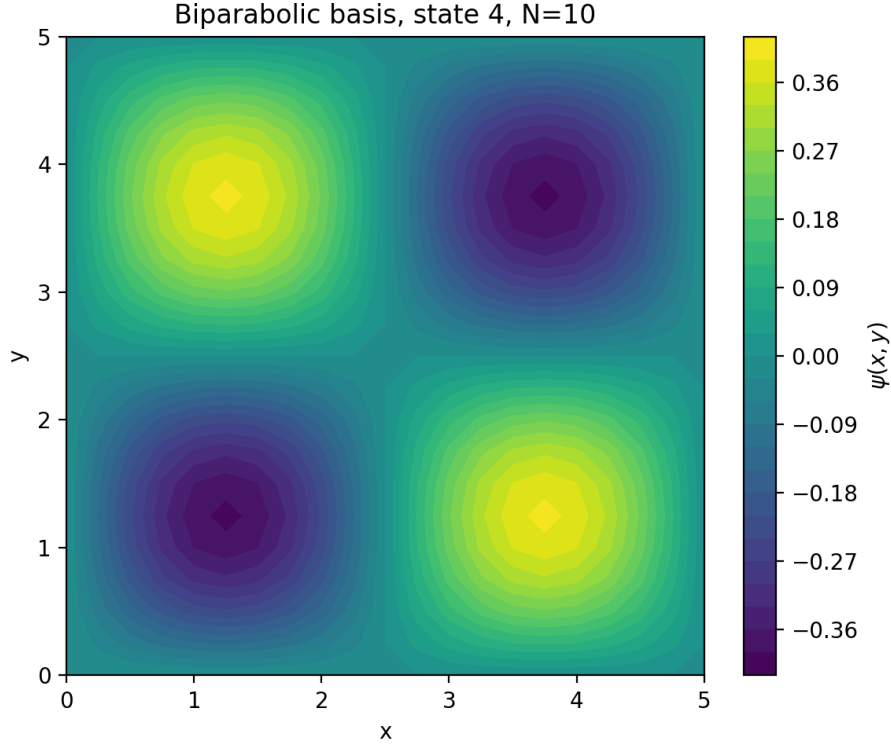


Figure 13: Wave function of the fourth positive eigenstate for the biparabolic basis, $N = 10$.

2.5 Summary of the numerical behavior

The obtained results show that the finite element Galerkin method correctly reproduces the low-energy spectrum of the two-dimensional infinite square well. For both basis types, the numerical eigenvalues converge toward the exact analytical energies as the mesh is refined. The convergence is monotonic in the sense that the discrepancy decreases with increasing N , and the absolute errors confirm this trend clearly.

A direct comparison of the two approximations demonstrates that the biparabolic basis is more accurate than the bilinear basis for the same mesh density. This improvement is visible both in the eigenvalue comparison and in the convergence plots. The wave functions also exhibit the expected nodal structures and symmetries of the exact eigenstates, confirming that the numerical method correctly captures not only the energy spectrum but also the spatial form of the eigenfunctions.

3 Conclusions

The finite element method in the Galerkin formulation correctly reproduces the low-energy spectrum of the two-dimensional infinite square well. In both cases, the numerical eigenvalues converge to the exact analytical results as the mesh is refined. The biparabolic basis gives higher accuracy than the bilinear basis for the same value of N .